Avnish Patel

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Publication

- B. Maxwell, A. Patel, "Logarithmic Lenses: Exploring Log RGB Data for Image Classification", CVPR 2024 [paper]
- S. Zhao, A. Patel, "MSO: Uncertainty-Aware Multi-Spectral Inertial Odometry for Extreme Environments", IROS 2026 Workshop [workshop] [website]

Experience

Medtronic | Robotics Software Engineer, Boston *Jan. 2024 – Present*

- Built an end-to-end *SLAM* pipeline using *stereo-rectified*, semantically segmented images in deep-learning based *DROID-SLAM* for *dense depth estimation* in surgical videos and scene understanding of anatomy and surgical tools
- Optimized *camera trajectory* using *GTSAM Factor Graphs non-linear optimization* and refining *3D reconstruction* with *Bundle Adjustment* and *Multi-View Geometry* with *LightGlue feature matching* and *Open3D offline rendering* pipeline
- Designed a *Pose Graph* pipeline with *GTSAM* using *Robot Kinematics* and *TensorRT* based *YOLOv8* vision keypoints to reduce the drift and improve *State Estimation* accuracy for Instrument tracking in *Out-of-View* scenarios
- Developed an *Oculus Quest*-based teleoperation pipeline in *NVIDIA Isaac Sim/Isaac Lab* for the *HUGO Surgical Robot* and trained a *PI-0.5 Vision-Language-Action (VLA)* model for autonomous tumor point-touching
- Trained a *PPO-based* reinforcement learning policy in *NVIDIA Isaac Lab* for automated surgical tumor point placement, leveraging differential *inverse kinematics* for precise and safe robotic motion control
- Engineered a *Arm Collision Avoidance System* using *Trimesh* and *Pytorch Volumetric Signed Distance Fields* of the medical robot using *URDF* and *Kinematics* using *RTI DDS publish-subcribe* method for rendering a *Digital Twin* of the robot
- *Deployed* a *C++* based *3D registration* pipeline on *NVIDIA Holoscan* using *Fast Foundation Stereo* and *UniMatch*, with custom *CUDA* kernels for high-performance *GPU* acceleration
- Engineered an *M3T* instrument tracking and articulation based framework in *C++* for real-time *6D pose* estimation
- Developed a *Monocular Depth Estimation* pipeline using *Depth Anything* to estimate real-time distances between surgical instrument tips, enabling spatial awareness and precise instrument interaction analysis
- Built a *MASt3R SFM*-based registration pipeline to create a *3D Digital Twin* of surgical robot carts using *GTSAM* for multi-camera scale correction and pose optimization, and *Signed Distance Fields (SDFs)* for collision avoidance.
- Developed a real-time *Ground Truth pose estimation pipeline* using *OptiTrack* Motion Capture system and *robot kinematics* for reconstructing and validating surigcal robot *Digital Twin* models
- Programmed a *Stereo-Vision and Stereo-Inertial localization* system using *ORB-SLAM3* and *calibrated RealSense D455* cameras, enabling accurate *ego-motion* tracking and mapping of *surgical cart trajectories* in a medical environment
- Automated *Endoscope Stereo Camera Calibration* using *Zhang's method* on a Charuco board with a *UR3 Robot Arm*
- Segmented hernias in 10,000 medical images using *Swin Base Transformer* with *PyTorch DDP* for *multi-GPU training*, tracked experiments via *MLflow*, and optimized using *ONNX/TensorRT*
- Implemented a *PyTorch* wrapper with *Optical Flow* on *FAST API* using *Unimatch*, deploying models to *ONNX/TensorRT* for 10*x* reduction in real time annotation of medical image frames with 1-*second* latency

Visiting Researcher - AirLab | Carnegie Mellon University, Remote *July 2025 – Aug. 2026*

- Executed *LiDAR-Inertial SLAM* and and *State Estimation* using *Livox LiDAR* and *IMU* sensors on the *Unitree G1 Humanoid* robot, applying a *low-pass filter* to mitigate *IMU bias* and enhance mapping accuracy
- Integrated *State Estimation* and *Relative Pose Graph Optimization* in *ROS2* in *C++* using *GTSAM Factor Graphs Fixed-Lag Smoother* in the *IMU Preintegration* module to reduce long term drift in *SLAM*
- Developed an *Active Bundle Adjustment* framework that jointly optimizes *sensor reliability amplitudes* and the *visual-inertial state* for *RGB*, *Thermal*, and *IMU* measurements, enabling adaptive sensor fusion
- Built a *Multi-Spectral Odometry* (*MSO*) system with *RGB/Thermal feature tracking* and a *sliding-window Bundle Adjustment* backend using *Ceres*, adapting sensor influence across *smoke*, *darkness*, and *low-texture* environments
- Evaluated *MSO* across *SubT-MRS*, *CART*, *EuRoC*, and *Wildfire* datasets, achieving a *27% reduction* in mean fused error across challenging visual conditions

Kisan Drip Irrigation Ltd. | Artificial Intelligence Engineer, India *Aug. 2020 – Aug. 2022*

- Integrated *ElasticFusion: RGB-D SLAM* with *C++* to align *multi-view point clouds* from *Intel RealSense D*455 *cameras*, enabling accurate *3D Reconstruction* for *pipe inspection* and *defect analysis*
- Experimented with *PointNet-based* deep learning models in *Python* for *point cloud classification* to enhance complex defect identification, achieving a 30% improvement over traditional 2D vision methods
- Deployed the *YOLO pipeline* as a *Docker containerized* FastAPI service integrated into manufacturing workflows

Education

Northeastern University | Boston, MA Sep. 2022 – Dec. 2024

Master of Science in Robotics, Specialization in Computer Vision